

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Pipeline MTO
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-LST-003
Document Revision	R01
Document Status	Issued For Review
Originator	Joseph Obinna
Security Classification	Restricted
Issue Date	2-Aug-22

Pipeline MTO

R01	02.08.22	Issued for Review	J. Obinna	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	02.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-LIST-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

Table of Contents

1. BBOU MTO	5
2. ELEPA MTO	6
3. AKABEL MTO	8
4. PIPELINE INDUCTION BEND LIST	9

3. MATERIAL TAKE OFF FOR PIPELINE BULKS - BBOU MANIFOLD														COMMENTS
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE	
PIPES														
1	PIPE (Mainline from BBOU and ELEPA)	ASME B31.8	24"	--	15.88mm	--	900#	ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	6,700.0	1000.0	7,700.0	Material is captured in LLI List
4	PIPE	ASME B31.8	24"	--	15.88mm	--	900#	ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	22.8	61.2	84	Material is captured in LLI List
2	PIPE	ASME B31.8	2"	--	80(5.54mm)	--	900#	API 5L /ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE X70 PSL 2,WT 5.54mm (12m SRL)	m	7.2	4.8	12	Material is captured in LLI List
3	PIPE	ASME B31.8	3/4"	--	160	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6	Material is captured in LLI List
TEES														
4	EQUAL BAR TEE	ASME B31.8	24"	24"	100	100	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	2	0	2	
WELDOLETS														
5	WELDOLET	ASME B31.8	24"	2"	100mm	160	900#	MSS-SP-98	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	7	1	8	
SOCKOLETS														
6	SOCKOLET	ASME B31.8	24"	3/4"	100	160	6000#	MSS-SP-98	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
ELBOW														
7	BLIND FLANGE	ASME B31.8	24"	--	24.2mm	--	900#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9,	PCS	2	0	2	
8	90 DEG. LR. ELBOW	ASME B31.8	2"	--	160	--	900#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9,	PCS	8	1	9	
FLANGES														
9	FLANGE	ASME B31.8	24"	--	100	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.5	PCS	12	1	13	
	FLANGE	ASME B31.8	2"	--	160	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.5	PCS	10	1	11	
10	FLANGE	ASME B31.8	3/4"	--	160	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#; RTJ; ASME B16.5	PCS	1	0	1	
11	BLIND FLANGE	ASME B31.8	3/4"	--	160	--	1500#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	1	0	1	
GASKETS														
12	GASKET	ASME B31.8	24"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	14	14	28	
13	GASKET	ASME B31.8	2"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	12	12	24	
14	GASKET	ASME B31.8	3/4"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	1	1	2	
STUDBOLTS & NUTS														
15	STUD BOLTS & HEX NUTS	ASME B31.8	2+1/2"	505mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	168	34	202	
16	STUD BOLTS & HEX NUTS	ASME B31.8	7/8"	160mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	96	19	115	
17	STUD BOLTS & HEX NUTS	ASME B31.8	3/4"	125mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	4	1	5	
SPECIAL PIPING ITEMS														
18	INSULATING JOINT	ASME B31.8	24"	--	--	--	900#	ASME 31.3 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	2	0	2	Material is captured in LLI List
4.0 MATERIAL TAKE OFF FOR PIPELINE VALVES -BBOU MANIFOLD														
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	BORE TYPE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO	COMMENTS
1	24" SDV 1002	ASME B31.8	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	
1	24"	ASME B31.8	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	5	0	5	Material is captured in LLI List
2	2"	ASME B31.8	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
3	3/4"	ASME B31.8	BALL	--		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#	--	SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	1	0	1	Material is captured in LLI List
4	2"	ASME B31.8	GLOBE	--		RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC,HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List

3. MATERIAL TAKE OFF FOR PIPELINE BULKS - ELEPA MANIFOLD															COMMENTS
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE		
PIPES															
1	PIPE (Mainline from Elapa to AKabel)	ASME B31.8	36"	--	23.83mm	--	900#	API 5L/ ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	31,500.0	1500.0	33,000	Material is captured in LLI List	
2	PIPE	ASME B31.8	36"	--	23.83mm	--	900#	API 5L/ ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	74.2	9.8	84	Material is captured in LLI List	
3	PIPE	ASME B31.8	2"	--	80/(5.54mm)	--	900#	API 5L/ ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE X70 PSL 2,WT 5.54mm (6m SRL)	m	13.3	10.7	24	Material is captured in LLI List	
4	PIPE	ASME B31.8	24"	--	15.88mm	--	900#	ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 15.88mm (12m SRL)	m	48.7	11.3	60	Material is captured in LLI List	
5	PIPE	ASME B31.8	3/4"	--	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6	Material is captured in LLI List	
TEES															
6	EQUAL BAR TEE	ASME B31.8	36"	36"	38.2mm	38.2mm	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	4	0	4		
7	EQUAL BAR TEE	ASME B31.8	24"	24"	100	100	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	1	0	1		
WELDOLETS															
8	WELDOLET	ASME B31.8	36"	2"	38.2mm	XXS	900#	MSS-SP-98	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	7	1	8		
9	WELDOLET	ASME B31.8	24"	2"	100	XXS	900#	MSS-SP-99	WELDOLET, MSS SP-97, ASTM A182 F52	PCS	2	0	2		
SOCKOLETS															
10	SOCKOLET	ASME B31.8	36"	3/4"	38.2mm	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	2	0	2		
11	SOCKOLET	ASME B31.8	24"	3/4"	100	XXS	6000#	MSS-SP-98	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1		
ELBOW															
12	90 DEG. LR. ELBOW	ASME B31.8	36"	--	38.2mm	--	900#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9,	PCS	2	0	2		
13	90 DEG. LR. ELBOW	ASME B31.8	2"	--	XXS	--	900#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9,	PCS	12	1	13		
FLANGES															
14	FLANGE	ASME B31.8	36"	--	38.2mm	--	900#	ASME B16.5	ASTM A182 F51, 'WN, 1500#, RTJ, ASME B16.47 SERIES A	PCS	14	1	15		
15	FLANGE	ASME B31.8	24"	--	100	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.5	PCS	3	0	3		
16	FLANGE	ASME B31.8	2"	--	XXS	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.5	PCS	13	1	14		
17	FLANGE	ASME B31.8	3/4"	--	XXS	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#; RTJ; ASME B16.5	PCS	3	0	3		
18	BLIND FLANGE	ASME B31.8	3/4"	--	XXS	--	1500#	ASME B16.5	'ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	3	0	3		
GASKETS															
19	GASKET	ASME B31.8	36"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	280	280	560		
20	GASKET	ASME B31.8	24"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	4	4	8		
21	GASKET	ASME B31.8	2"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	14	14	28		
22	GASKET	ASME B31.8	3/4"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	3	3	6		
STUDBOLTS & NUTS															
23	STUD BOLTS & HEX NUTS	ASME B31.8	3+1/2"	605mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	60	12	72		
24	STUD BOLTS & HEX NUTS	ASME B31.8	2+1/2"	505mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	80	16	96		
25	STUD BOLTS & HEX NUTS	ASME B31.8	7/8"	160mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	112	22	134		
26	STUD BOLTS & HEX NUTS	ASME B31.8	3/4"	125mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	12	2	14		
SPECIAL PIPING ITEMS															
27	INSULATING JOINT	ASME B31.8	36"	--	--	--	900#	ASME 31.8 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	2	0	2		
28	INSULATING JOINT	ASME B31.8	24"	--	--	--	900#	ASME 31.3 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	1	0	1		

4.0 MATERIAL TAKE OFF FOR PIPELINE VALVES - ELEPA														COMMENTS
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE	
29	36" SDV1002	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
30	36"	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
31	24" SDV 1003	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
32	24"	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
33	2"	ASME B31.8	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR	PCS	4	0	4	Material is captured in LLI List
34	3/4"	ASME B31.8	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#	—	SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	3	0	3	Material is captured in LLI List
35	2"	ASME B31.8	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SW/VEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	3	0	3	Material is captured in LLI List

3. MATERIAL TAKE OFF FOR PIPELINE BULKS - AKABEL RECEIVER														COMMENTS
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE	
PIPES														
1	PIPE	ASME B31.8	36"	--	23.83mm	--	900#	API 5L / ASME B36.10M	API 5L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (2m SRL)	m	33.8	2.2	36	Material is captured in LLI List
2	PIPE	ASME B31.8	2"	--	80(5.54mm)	--	900#	API 5L / ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE X70 PSL 2, WT 5.54mm (6m SRL)	m	4.7	19.3	24	Material is captured in LLI List
14	PIPE	ASME B31.8	3/4"	--	XXS	--	900#	API 5L / ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6	Material is captured in LLI List
TEES														
15	EQUAL BAR TEE	ASME B31.8	36"	36"	23.83mm	23.83mm	900#	ASME B16.9	ASTM A890 WPHY 60, BW, ASME B16.9	PCS	3	0	3	
WELDOLETS														
27	WELDOLET	ASME B31.8	36"	2"	38.2mm	XXS	900#	MSS-SP-98	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	9	1	10	
SOCKOLETS														
38	SOCKOLET	ASME B31.8	36"	1 1/2"	38.2mm	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	2	0	2	
	SOCKOLET	ASME B31.8	36"	3/4"	38.2mm	XXS	6000#	MSS-SP-98	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
ELBOW														
47	90 DEG. LR. ELBOW	ASME B31.8	36"	--	38.2mm	--	900#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9	PCS	6	1	7	
48	90 DEG. LR. ELBOW	ASME B31.8	2"	--	XXS	--	900#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9	PCS	4	0	4	
FLANGES														
60	FLANGE	ASME B31.8	36"	--	38.2mm	--	900#	ASME B16.5	ASTM A182 F51, WN, 1500#, RTJ, ASME B16.47 SERIES A	PCS	9	1	10	
	FLANGE	ASME B31.8	2"	--	XXS	--	900#	ASME B16.5	ASTM A694 F80, WN, 900#, RTJ, ASME B16.5	PCS	13	1	14	
71	FLANGE	ASME B31.8	3/4"	--	XXS	--	1500#	ASME B16.5	ASTM A105N; SW: 1500#; RTJ; ASME B16.5	PCS	1	0	1	
	BLIND FLANGE	ASME B31.8	3/4"	--	XXS	--	1500#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	1	0	1	
73	REDUCING FLANGE	ASME B31.8	2"	1 1/2"	XXS	XXS	1500#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	1	0	1	
GASKETS														
77	GASKET	ASME B31.8	36"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	11	11	22	
78	GASKET	ASME B31.8	2"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	12	12	24	
	GASKET	ASME B31.8	1 1/2"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	2	2	4	
87	GASKET	ASME B31.8	3/4"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	3	3	6	
STUDBOLTS & NUTS														
91	STUD BOLTS & HEX NUTS	ASME B31.8	3+1/2"	605mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DP GALVANIZED (HDG) PER ASTM A153	PCS	220	44	264	
92	STUD BOLTS & HEX NUTS	ASME B31.8	7/8"	160mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DP GALVANIZED (HDG) PER ASTM A153	PCS	240	48	288	
	STUD BOLTS & HEX NUTS	ASME B31.8	1"	155mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DP GALVANIZED (HDG) PER ASTM A153	PCS	8	2	10	
101	STUD BOLTS & HEX NUTS	ASME B31.8	3/4"	125mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DP GALVANIZED (HDG) PER ASTM A153	PCS	12	2	14	
SPECIAL PIPING ITEMS														
107	INSULATING JOINT	ASME B31.8	36"	--	--	--	900#	ASME 21.8 ASME VIII DIV.	BODY: API 5L GR.B; PUPS: API 5L GR.B; PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	1	0	1	Material is captured in LLI List
4.0 MATERIAL TAKE OFF FOR PIPELINE VALVES - AKABEL RECEIVER														
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO	COMMENTS
1	36"	ASME B31.8	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASYM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, INDOVHL, OPR, W/ GEARBOX	PCS	4	0	4	
8	2"	ASME B31.8	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASYM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR	PCS	4	0	4	Material is captured in LLI List
11	3/4"	ASME B31.8	BALL	--	--	SW WITH 100mm PUP PIECE	1500#	--	ASYM A105N BODY, SS316 TRIM, FLOATING BALL, LEVER OPR	PCS	1	0	1	Material is captured in LLI List
14	2"	ASME B31.8	GLOBE	--	--	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC INDOVHL, OPR, W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List

INDUCTION BEND LIST PIPELINE FROM BUBOUWE MANIFOLD STATION TO ELEPA MANIFOLD STATION										COMMENTS
SN	PIPE ND	ANGLE (DEGRE E)	MATERIAL	WALL THICKNE SS (mm)	COATING/ THICKNESS	KP (Km)	NORTHING (m)	EASTING (m)	LOCATION	
1	24"	30	API 5L GR X70	15.88	3LPE/3mm	0+000	494208.764	179065.951	BUBOUWE MANIFOLD STATION	Material is captured in LLI List
2	24"	30	API 5L GR X70	15.88	3LPE/3mm	0+000	494209.287	179066.566	BUBOUWE MANIFOLD STATION	Material is captured in LLI List
3	24"	75	API 5L GR X70	15.88	3LPE/3mm	0+251	494371.818	179257.006	INFIELD	Material is captured in LLI List
4	24"	90	API 5L GR X70	15.88	3LPE/3mm	6+988	490530.889	184791.027	INFIELD	Material is captured in LLI List
5	24"	35	API 5L GR X70	15.88	3LPE/3mm	7+067	490596.545	184836.595	INFIELD	Material is captured in LLI List
6	24"	90	API 5L GR X70	15.88	3LPE/3mm	7+314	490842.142	184835.702	INFIELD	Material is captured in LLI List
7	24"	30	API 5L GR X70	15.88	3LPE/3mm	7+321	490843.035	184818.54	ELEPA MANIFOLD STATION	Material is captured in LLI List
8	24"	30	API 5L GR X70	15.88	3LPE/3mm	7+321	490843.037	184826.388	ELEPA MANIFOLD STATION	Material is captured in LLI List

INDUCTION BEND LIST PIPELINE FROM ELEPA MANIFOLD STATION TO AKABEL MANIFOLD STATION										COMMENTS
SN	PIPE ND	ANGLE (DEGRE E)	MATERIAL	WALL THICKNE SS (mm)	COATING/ THICKNESS	KP (Km)	NORTHING (m)	EASTING (m)	LOCATION	
1	36"	30	API 5L GR X70	23.83	3LPE/3mm	0+000	490847.441	184828.562	ELEPA MANIFOLD STATION	Material is captured in LLI List
2	36"	30	API 5L GR X70	23.83	3LPE/3mm	0+000	490847.406	184834.527	ELEPA MANIFOLD STATION	Material is captured in LLI List
3	36"	90	API 5L GR X70	23.83	3LPE/3mm	0+010	490847.387	184839.899	INFIELD	Material is captured in LLI List
4	36"	55	API 5L GR X70	23.83	3LPE/3mm	0+361	490496.969	184839.899	INFIELD	Material is captured in LLI List
5	36"	34	API 5L GR X70	23.83	3LPE/3mm	3+581	488660.921	187485.282	INFIELD	Material is captured in LLI List
6	36"	51	API 5L GR X70	23.83	3LPE/3mm	14+855	488421.13	198756.58	INFIELD	Material is captured in LLI List
7	36"	34	API 5L GR X70	23.83	3LPE/3mm	19+356	484882.923	201538.782	INFIELD	Material is captured in LLI List
8	36"	32	API 5L GR X70	23.83	3LPE/3mm	25+227	483106.542	207134.625	INFIELD	Material is captured in LLI List
9	36"	30	API 5L GR X70	23.83	3LPE/3mm	31+376	478456.149	211155.233	AKABEL MANIFOLD STATION	Material is captured in LLI List
10	36"	30	API 5L GR X70	23.83	3LPE/3mm	31+376	478451.284	211159.476	AKABEL MANIFOLD STATION	Material is captured in LLI List